

CHINA MATTERS

From Weak to Weaker

- China's July activity data showed renewed growth weakness, and we estimate that official real GDP is running at around 4% yoy in early Q3. The combination of lower volumes and lower prices indicates that the weakness is demand-driven.
- July's growth deceleration is more concerning than April's because it came from a lower starting point and impacted areas that had previously looked resilient. Q2 real GDP growth was only 4.3% yoy, below the government's full-year target of 4.5–5%. By our estimates, growth of services retail sales slowed to 3.2% yoy in July after running above 5% earlier this year, raising questions about household consumption momentum.
- Looking back over the past few years, policy support has been too small relative to the shock, leaving the private sector under sustained financial pressure. Policymakers' near-exclusive focus on technology innovation and high-tech manufacturing does not easily lend itself to a more virtuous cycle of stronger employment, income and consumption, as manufacturing accounts for only around 20% of employment in China. These factors help explain why growth tends to soften notably whenever headwinds emerge.
- After the July Politburo meeting turned more dovish, the government announced various easing measures, including faster government bond issuance and starting the new policy-based financing instrument to invest in the "Six Networks" and major projects in the 15th Five-Year Plan, support for large sectors with long value chains (likely electronic equipment and machinery), relaxed Housing Provident Fund rules, and measures to boost consumption in lower-tier cities and rural areas.
- Upcoming measures may help the government meet this year's growth target, but they remain mostly supply-driven and are unlikely to create durable momentum. In our view, without stronger demand-side stimulus and fundamental reforms to unleash domestic demand, cyclical weakness risks becoming structural hysteresis.

Hui Shan

+852-2978-6634 | hui.shan@gs.com
Goldman Sachs (Asia) L.L.C.

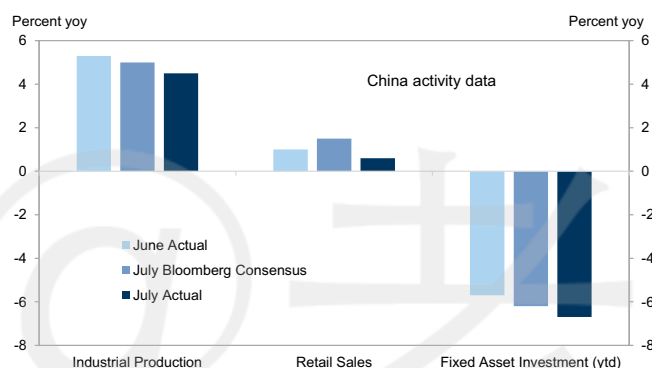
From Weak to Weaker

A Weak Batch of Data

China's July activity data weakened from June and disappointed already-low market expectations. Industrial production (IP) was relatively resilient, rising 4.5% yoy, supported by robust export growth. By contrast, retail sales grew only 0.6% yoy, and fixed asset investment (FAI) contracted 6.7% yoy in the first seven months of this year, implying double-digit year-over-year declines in July. With the construction sector still contracting and high-frequency data not pointing to a noticeable pickup, the July official IP and services output index suggest that China's official real GDP growth is running at a pace of around 4% yoy in the first half of Q3.¹

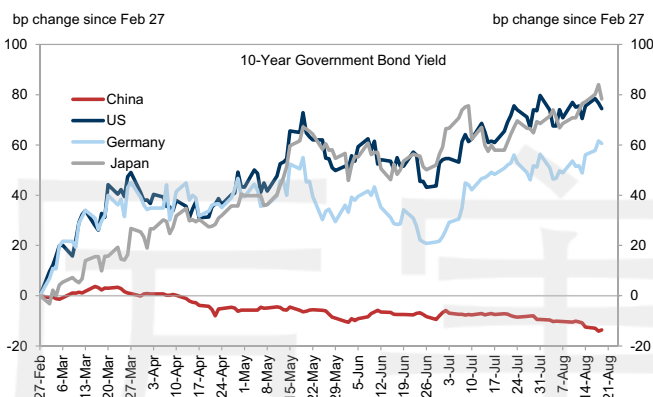
Following the weaker-than-expected macro data and the PBOC's mid-month overnight reverse repo operations to inject liquidity, our conversations with traders and investors suggest that market expectations of monetary policy easing increased somewhat. The yield on 10-year Chinese Government Bonds (CGB) also fell below 1.70%. Compared with end-February, just before the start of the Iran War, 10-year CGB yields have declined by more than 10bp, suggesting the market's view that this year's energy-driven inflation is likely temporary, growth headwinds remain strong, and policy easing is warranted. In sharp contrast, yields on 10-year government bonds in other countries have climbed substantially since late February (Exhibit 2).

Exhibit 1: July activity data weakened from June and missed market expectations



Source: Bloomberg, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 2: Yields on 10-year governments climbed in other countries but declined in China



Source: Bloomberg, Goldman Sachs Global Investment Research

Clear Evidence of Demand-Driven Weakness

Recent data from the steel sector provide clear evidence that weak demand is the key driver of sluggish economic activity in China. Weekly steel production data show falling output in July and early August (Exhibit 3). At the same time, steel prices—especially rebar prices, which are mainly used in construction—have declined in recent months (Exhibit 4). The combination of lower volumes and lower prices points to weak demand

¹ Note that China's fixed asset investment (FAI) series has significantly diverged from the gross capital formation (GCF) series under GDP. In the first half of 2026, for example, FAI contracted by 5.7% yoy whereas GCF still contributed 1.8pp to year-over-year real GDP growth in the official data. See our research here for more details about this divergence.

as the cause of slowing economic growth.

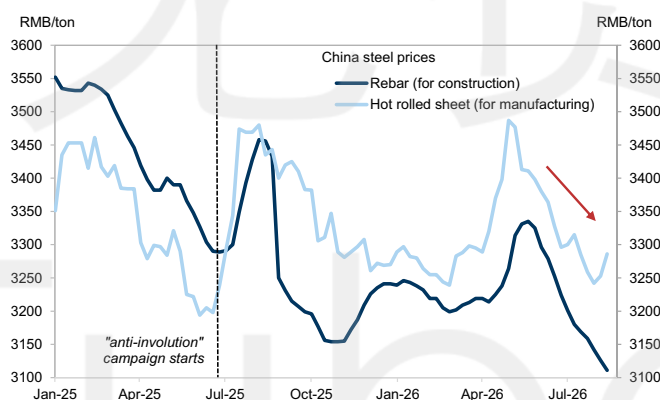
The “anti-involution” campaign that started last July led to sharply higher prices in a few sectors, such as steel and polysilicon, for several months. However, without solid demand to support higher prices, and with concerns over labor market stability preventing significant supply cuts, prices eventually fell back to levels similar to, or even below, those seen before the campaign.

Exhibit 3: Steel production dropped in July and early August



Source: Mysteel, Goldman Sachs Global Investment Research

Exhibit 4: Steel prices have declined in recent months



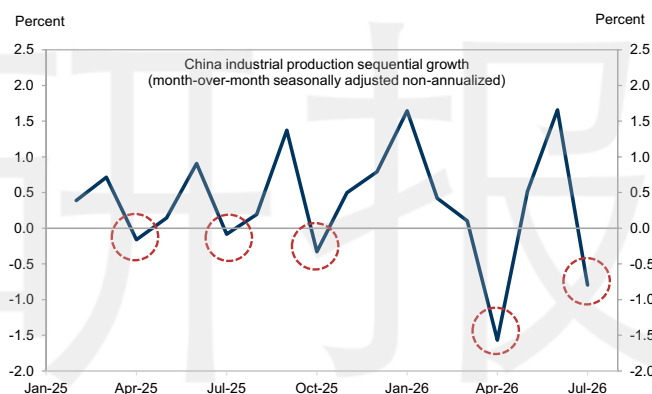
Source: Bloomberg, Goldman Sachs Global Investment Research

Why We Are More Concerned About July Data Weakness

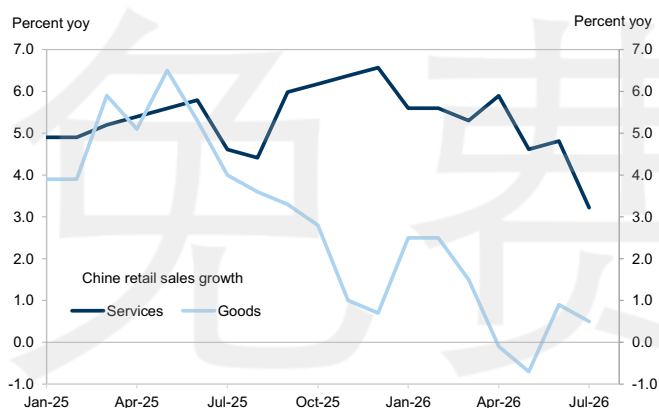
Over the past year, we have observed a consistent pattern of IP and the NBS manufacturing PMI strengthening into quarter-end months, such as March, June, and September, before falling sequentially in subsequent months, such as April, July, and October ([Exhibit 5](#)). In our view, this pattern may reflect manufacturing output being frontloaded into quarter-end months to support stronger readings of production-based official GDP measures. Back in April, we were not too concerned about the sequential decline in growth momentum. However, we think the July weakness points to increased downside risks for three reasons.

First, the April growth deceleration came after 5.0% yoy real GDP growth in Q1, which was at the upper end of this year's 4.5–5% growth target range. By contrast, the July growth deceleration followed 4.3% yoy real GDP growth in Q2, which was below the lower bound of the range.

Second, some sectors that had appeared organically strong showed fading momentum in July. For example, within retail sales, services sales growth had been running above 5% yoy earlier this year, significantly outperforming goods sales. In July, however, services sales growth dropped to 3.2% yoy by our estimates, raising questions about the outlook for household consumption, as goods sales are expected to remain soft amid the fading boost from the consumer goods trade-in program ([Exhibit 6](#)).

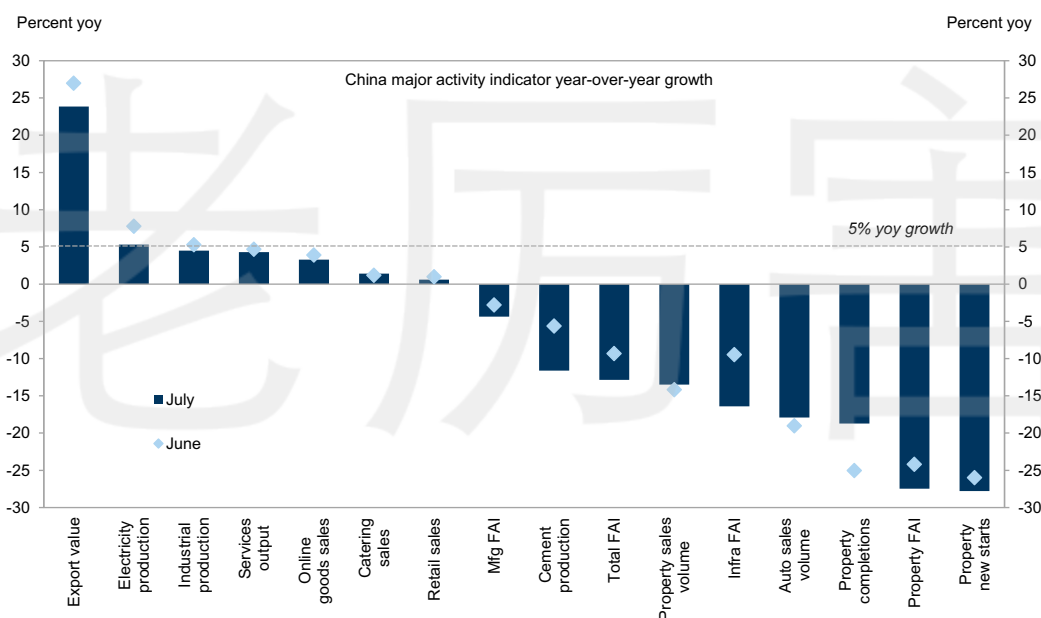
Exhibit 5: IP growth tends to decelerate sharply in quarter-start months

Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 6: Retail sales of services decelerated notably in July

Source: Wind, Goldman Sachs Global Investment Research

Third, some of the headwinds present in April turned out to be more persistent than we initially anticipated. Adverse weather conditions continued through August and may linger into year-end due to El Niño effects. Local government incentives are lacking in a political turnover year, while the number of anti-corruption investigations continues to reach new highs. These factors have contributed to slow government bond issuance, even as growth pressures increase and unused issuance quotas remain abundant. Among major indicators, year-over-year growth in infrastructure FAI and cement output declined the most from June to July ([Exhibit 7](#)).

Exhibit 7: Year-over-year growth in infrastructure FAI and cement production declined the most from June to July

Source: CEIC, Goldman Sachs Global Investment Research

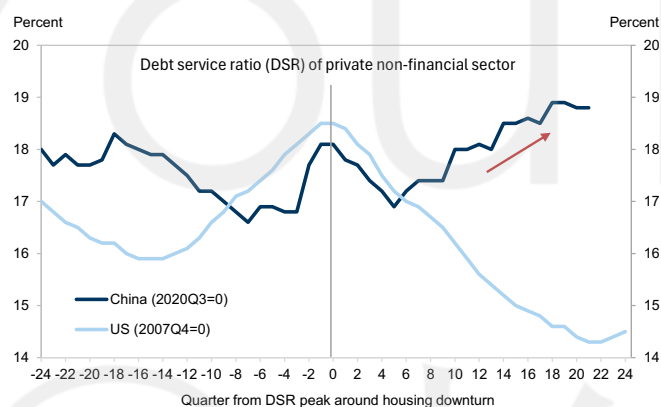
Causes of the Persistent Weakness

One fundamental reason why the domestic economy has softened again, despite the

easing package announced in September 2024 and the RMB10tn debt resolution plan approved in November 2024, is that policy support over the past few years has been too small relative to the size of the shock. One example is the evolution of the debt-service ratio (DSR) of the private non-financial sector. According to BIS data, the US private non-financial sector DSR peaked in 2007Q4 at the start of the subprime crisis (Exhibit 8). Significant Fed rate cuts and various government debt restructuring programs helped reduce the DSR by more than 4 percentage points over the ensuing seven years.

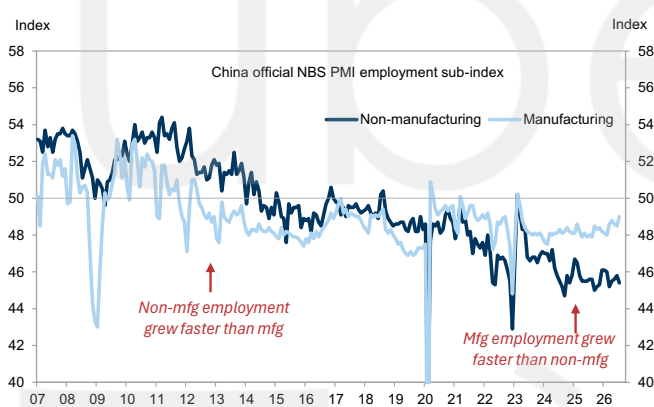
In contrast, China's private non-financial sector DSR initially peaked in 2020Q3, when the “three red lines” policy aimed at deleveraging the property market was first introduced. But later, between 2022Q1 and 2025Q4, China's DSR increased by almost 2 percentage points. In other words, against the backdrop of three years of Covid lockdowns and a severe housing downturn, China's private sector had to allocate a rising share of income to debt servicing over the past five years. These elevated financial burdens squeezed purchasing power, weighed on sentiment, and left the domestic economy prone to notable weakening whenever growth headwinds emerged.

Exhibit 8: Unlike the US experience during the subprime crisis, China's private sector debt-service ratio rose during the housing downturn



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 9: Manufacturing employment has outperformed non-manufacturing employment in recent years, reversing the pre-Covid trend



Source: Haver Analytics, Goldman Sachs Global Investment Research

Another fundamental reason lies in the government's strategy of focusing almost exclusively on tech innovation and high-tech manufacturing. Before the Covid pandemic, the employment subindex of the NBS non-manufacturing PMI significantly outperformed that of the NBS manufacturing PMI. Post-Covid, however, this relationship completely reversed (Exhibit 9). While manufacturing employment appears better supported in recent years, it accounts for only around 20% of China's total employment and is insufficient to offset weakness in non-manufacturing employment. Put differently, while the policy emphasis on tech innovation and high-tech manufacturing has led to breakthroughs such as DeepSeek, as well as increased manufacturing competitiveness and persistent export strength, it has not translated into broad-based employment and income growth, which are crucial to a sustained increase in domestic demand.

Policy Plans in Pipeline and Their Limits

To be clear, despite the weak July activity data and the downside risk to our Q3 GDP forecast, we still believe the government will manage to achieve at least 4.5% official real GDP growth this year and meet its target, barring extraordinary external shocks. After all, there have been only two instances when the government missed its growth target:

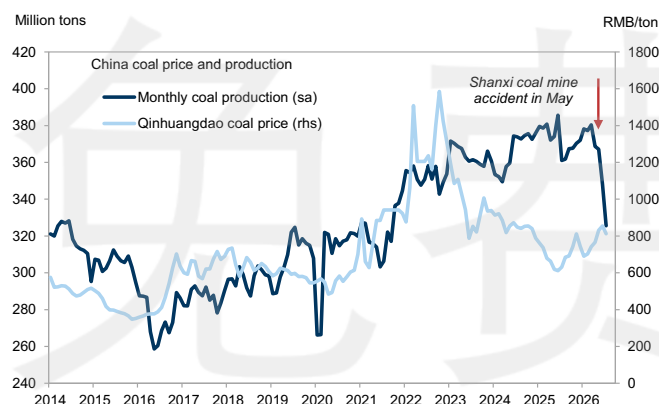
once in 1998 during the Asian Financial Crisis, and again in 2022, when the spread of the Omicron variant proved too difficult to contain. The July Politburo meeting already turned more dovish by calling for “*strengthening countercyclical measures*.” Various official announcements and government meetings after the July Politburo meeting point to four types of policies in the remainder of the year to support GDP growth (Exhibit 10).

- **Investment:** As of mid-August, nearly half of the full-year government bond issuance quota remained untapped. To our knowledge, the RMB800bn New Policy-Based Financing Instrument (NPBFI) allocated for 2026 had not been used either. The first step would therefore be to accelerate government bond issuance and kick-start the NPBFI program. Targeted areas of spending are likely to include the “Six Networks” (i.e., water networks, new power grids, computing power networks, next-generation communication networks, urban underground pipeline networks, and logistics networks) and the major projects outlined in the 15th Five-Year Plan.
- **Industry:** During the 12th State Council plenary session on August 17, Premier Li Qiang called for a focus on “*sectors characterized by long value chains and large scale*” to boost domestic demand. Although property, autos, home appliances, electronic equipment and machinery all fit this description, the first three appear unlikely to see major new policies stimulating production, as property destocking is still ongoing and auto and home appliance sales were already frontloaded by the government-subsidized consumer goods trade-in program. This leaves electronic equipment and machinery, including semiconductors, as the likely sectors to receive additional policy support to expand production.
- **Housing:** On August 18, the State Council announced changes to the Housing Provident Fund (HPF) rules to expand its use. The HPF is a mandatory, government-backed savings scheme in which employers and employees both contribute to individual accounts for housing-related consumption, such as buying, renting, or renovating a home. Currently, interest rates on HPF loans are 50bp lower than mortgage rates for first-time homebuyers. However, we think the impact of the latest announcement will only have a marginal impact on the property market, as many cities have already relaxed local HPF rules, benefiting low-end properties.
- **Consumption:** On August 13, the Ministry of Commerce, along with eight other ministries, released a document on stimulating consumption in lower-tier cities and rural areas. Policymakers regard these areas as having greater demand for consumption upgrading. However, many of the measures focus on the supply side, including renovating commercial facilities and expanding the supply of high-quality, premium-priced goods. Without bolstering income, overall consumption is unlikely to rise simply because of increased supply of goods and services.

Exhibit 10: Policy plans to boost growth in the remainder of the year

Type	Policy plans based on official statements
Investment	Accelerating government bond issuance; starting to use new policy-based financial instrument; "Six Networks" construction and major projects outlined in the 15th Five-Year Plan
Industry	Large sectors with long value chains (potentially electronics and equipment/machinery, including semiconductors)
Housing	Expanding and easing the use of housing provident fund (HPF)
Consumption	Boosting consumption in lower-tier cities and rural areas

Source: Government websites, Goldman Sachs Global Investment Research

Exhibit 11: Coal production declined sharply after the Shanxi coal mine accident in May

Source: Bloomberg, CEIC, Goldman Sachs Global Investment Research

It remains to be seen how effective these policies will be, and whether policymakers will need to roll out additional easing measures in October (after seeing Q3 real GDP growth) in order to ensure the full-year growth target is met. In recent months, the government has introduced several safety- and security-related measures, including a special campaign against organized crime and tightened border control regulations. While safety and security need to be protected, these policies, especially when aggressively implemented, could have a negative impact on economic activity. For example, after the Shanxi coal mine accident in May, Chinese coal production dropped 12%, similar to the decline in January–February 2020 when the country was under a national Covid lockdown (Exhibit 11). In the current environment, where top leadership and local officials place strong emphasis on safety and security, the transmission from policy easing to economic activity may be blunted.

Even if the cyclical measures listed in Exhibit 10 help the government achieve the GDP growth target this year, they are unlikely to generate sustained growth momentum. This package of measures appears mostly supply-driven, raising output and production-based GDP measures in the near term but often resulting in oversupply, deflation, and increased debt over time. Another risk from the continued lack of demand-side stimulus and fundamental reforms to unleash domestic demand is that cyclical weakness could turn into structural hysteresis. As discussed earlier, we believe the magnitude of policy easing over the past few years has been too small relative to the size of the shock, and the cautious, intermittent style of policy response appears to persist. If continued for a few more years, prolonged financial stress on households, private businesses, and local governments could impair future growth potential.

Hui Shan**The China Economics Team**

Andrew Tilton
+852-2978-1802
andrew.tilton@gs.com
Goldman Sachs (Asia) L.L.C.

Hui Shan
+852-2978-6634
hui.shan@gs.com
Goldman Sachs (Asia) L.L.C.

Lisheng Wang
+852-3966-4004
lisheng.wang@gs.com
Goldman Sachs (Asia) L.L.C.

Xinquan Chen
+852-2978-2418
xinquan.chen@gs.com
Goldman Sachs (Asia) L.L.C.

Yuting Yang
+852-2978-7283
yuting.yang@gs.com
Goldman Sachs (Asia) L.L.C.

Chelsea Song
+852-2978-0106
chelsea.song@gs.com
Goldman Sachs (Asia) L.L.C.

研报免费
YouTube
@老厉害

Disclosure Appendix

Reg AC

I, Hui Shan, hereby certify that all of the views expressed in this report accurately reflect my personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Hui Shan Goldman Sachs (Asia) L.L.C..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are

available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.